HW 7

To be completed by Wednesday April 8

1. Use "dataHW7 Problem1.mat" to complete this exercise. The data is identical to "dataHW6 Problem2.mat" - meaning the structure of the problem is the same as are the underlying parameters. The purpose of this problem is to estimate a simple dynamic discrete choice problem using CCPs.

   The data contain information on the choices (the Y's), characteristics that do not vary over time (the X_1's), and characteristics that do vary over time (the X_{1t}'s). The context of the problem is a dynamic model of labor supply. Y takes values $\{0, 1, 2, 3\}$ reflecting retirement, flex-time, part-time, or full-time participation. The first column X_1 allows for choice specific constants, while the second column of X_1 is an indicator for gender. X_{1t} is time-varying measure of health.

   Both sets of X's are individual-level characteristics so the coefficients on these variables need to vary by choice. There is a cost to switching labor supply in the future that is *common across all switches*. The choice in period 0 is given by LY. Once an individual retires (or chooses option 0) then that individual makes no further choices (you can verify this as if $Y_{it} = 0$ then $Y_{it+1} = 0$). The columns of Y and X_{1t} indicate observations over time, ten periods in all (so in period 10 it is a static problem). Individuals are uncertain about future health. Given X_{1t} and Y_t, X_{1t+1} is given by:

   $$X_{1t+1} = \alpha_0 + \alpha_1 X_{1t} + \alpha_2(Y_t == 2) + \alpha_3(Y_t == 3) + \epsilon_{t+1}$$

   for $Y_t > 0$ where ϵ_{t+1} is distributed $N(0, \sigma)$ and is unknown to the individual until time $t+1$.

   (a) Estimate the transition parameters, the α's and standard deviation of ϵ. Use these parameters to construct transition matrices.

   (b) Estimate the CCPs. To do this estimate a single logit model of retirement based on the state variables (X_1, X_{1t}, LY) and a set of dummy variables that indicate the time period. Once someone retires they should not contribute further to the likelihood.

   (c) Using the estimates from the first two steps, construct the future value terms associated with each choice for every individual in every period. This is a weighted average of the log of the probability of retiring in the next period where the weights are determined by the transition probabilities.

   (d) Finally, estimate the choice specific parameters, switching cost, and discount factor using a static multinomial logit. The future value terms derived in part (c) are simply additional regressors beyond X_1, X_{1t}, and LY.

True params

- Coefficients on X's (constant, gender, health) for three choices $[-.75; .25; .25][-0.25; 0.5; .5][-.85; .75; .75]$
- Switching cost: 0.4 (enters negatively)
- Discount factor: 0.8